

Canadian Civic Infrastructure Alliance

Submission on Ontario's draft Data Centre Playbook

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A multi-municipality public-interest submission on local authority, enforceable standards, cost recovery, transparency, economic benefits, and digital sovereignty.

About us

The Canadian Civic Infrastructure Alliance is a Canadian not-for-profit organization working with residents, municipal officials, researchers, and policy experts to strengthen public capacity around major digital and energy infrastructure decisions.

Our current work on data-centre development involves communities and municipal actors across Ontario and the country. While specific local circumstances can differ, the underlying governance issue is the same. Municipalities are being asked to assess large infrastructure projects of a novel nature even while existing standards and frameworks remain non-enforceable, incomplete, and outdated. Project information is often fragmented or opaque, making local costs and benefits difficult to assess and verify.

We welcome Ontario's decision to develop a provincial Data Centre Playbook. For the framework to succeed, it must treat municipalities as governments with independent authority over host-community impacts, not as mere stakeholders to be consulted only after major investment and grid decisions have already been made.

Summary

Ontario's final Data Centre Playbook should:

- require formal support from host municipalities for major projects
- give substantive weight to municipal input during project evaluation
- establish enforceable standards for local impacts including noise, water, and backup-energy generation
- ensure full recovery of municipal and electricity-system costs
- convert claimed economic and community benefits into measurable commitments, with strong disincentives for failing to follow through
- require transparency around ownership, data governance, lobbying, NDAs and project assessments
- apply a more rigorous test of digital sovereignty than physical data residency alone
- require 'behind-the-meter' power to be generated using renewable sources
- apply to facilities operating at the scale of 10MW and above

1. Municipalities need formal decisionmaking authority

Ontario's draft framework contemplates significant provincial control over which large-load projects receive access to the electricity system.

Municipalities must have a meaningful and clearly defined role in this process.

These projects place significant demands on municipalities. Considerations include land-use compatibility, roads, servicing capacity, emergency response, water and wastewater infrastructure, local noise, and community impacts.

The final framework should therefore require a formal resolution of support from the host municipality before a major data-centre project receives provincial grid approval or comparable provincial facilitation. Notably, Kincardine staff have also proposed that provincial grid approval be conditional on formal municipal support.

Across Ontario, municipalities are increasingly confronted with data-centre proposals, but lack local planning frameworks updated for this new class and scale of infrastructure. In many communities, councils are considering or have adopted interim controls precisely because existing zoning categories were never designed to distinguish modern high-intensity computing facilities from more conventional industrial or commercial uses.

A provincial framework should strengthen local decision-making, taking care not to inadvertently bypass it.

Recommendation: Require formal host-municipality support as a condition of provincial approval for major regulated data-centre projects.

2. Municipal input should carry substantive weight in project evaluation

The Province's assessment process should not treat municipal participation as a procedural checkbox.

Host municipalities are best positioned to evaluate:

- land-use compatibility
- road and infrastructure requirements
- water and wastewater capacity
- emergency-service needs
- housing and workforce pressures
- local environmental impacts
- community priorities

The final Playbook should specify when municipal input occurs, what information municipalities receive, and how that input affects the final decision.

An appropriate municipal role requires sufficient information and authority. Municipal governments cannot evaluate a project responsibly if material details about electrical demand, cooling systems, backup generation, ownership, operating structure or infrastructure requirements remain unavailable.

Recommendation: Make municipal input a mandatory component of project evaluation and require the Province to demonstrate how host-municipality concerns were considered in the final decision.

3. Ontario should establish enforceable local-impact standards

One of the clearest lessons emerging across municipalities is that voluntary commitments to "responsible development" are not enough.

Communities need measurable and binding standards.

Noise

Data centres operate continuously and can produce persistent mechanical and low-frequency noise from cooling equipment, electrical infrastructure, and backup-generation systems.

The Playbook should establish enforceable requirements concerning:

- continuous and low-frequency noise, including frequencies below the audible spectrum
- appropriate thresholds
- setbacks
- baseline and operational monitoring
- complaint investigation
- operator-funded compliance reporting

Technological guidance such as for acoustic enclosures or quieter cooling systems is useful, but design recommendations are not substitutes for standards.

Water

Proponents should disclose anticipated water consumption, cooling-system design, wastewater implications and contingency requirements.

Water-efficient and closed-loop systems should be required where appropriate, but the Province should also consider cumulative effects on municipal systems and local watersheds.

Backup and behind-the-meter generation

Grid-connection costs are only one part of a facility's energy footprint.

The final Playbook should establish standards for diesel, natural-gas, and other backup or behind-the-meter generation, including emissions reporting and operating limits.

Recommendation: Establish measurable provincial minimum standards for noise, water, backup generation, and other local environmental impacts while preserving municipalities' ability to adopt stronger local requirements when appropriate.

4. Host communities should not subsidize projects through unrecovered costs

Ontario has correctly emphasized that large data centres should bear the electricity-system costs associated with serving them. This should also apply municipally.

Proponents should demonstrate that host municipalities will not incur unrecovered costs arising from:

- roads and transportation
- water and wastewater impacts
- additional emergency services capacity requirements
- planning and building services
- municipal utility upgrades
- infrastructure expansion
- other growth-related requirements

These fiscal questions should be answered before approval, not after communities are locked into new, unavoidable infrastructure demands.

Recommendation: Require full municipal cost recovery as part of the provincial assessment process.

5. Economic and community benefits should be measurable

Large capital-investment figures can create an impression of extraordinary local economic benefit while obscuring how much of that spending flows to imported computing equipment and hyper-specialized contractors.

Ontario should distinguish between:

- construction employment
- permanent operational employment
- local procurement
- international supply-chain spending
- benefits actually retained within Ontario communities

Where proponents claim benefits such as local employment, training, compute access for Ontario businesses or infrastructure investment, those claims should become measurable project commitments.

Facilities should submit Community Benefit Plans or Agreements with clear metrics and annual reporting. Potential measures could include:

- permanent local jobs
- apprenticeships and training, by number of spots
- Local- or domestic-first procurement
- partnerships with colleges and universities
- municipal infrastructure contributions
- community investment
- Affordable low-latency compute access where this is presented as a project benefit
- Ineligibility for municipal tax subsidies and economic development incentives

Benefits should also be benchmarked against project characteristics such as electrical demand, land area, capital investment and municipal servicing impacts.

Recommendation: Require measurable Community Benefit Plans and distinguish permanent economic benefits from temporary construction activity.

6. Transparency must be built into the framework

Ontario's final framework should require sufficient public disclosure regarding:

- beneficial and operating ownership
- facility operators
- cloud or compute providers
- material project partners
- electrical demand
- water use
- backup generation
- community-benefit commitments
- meetings between proponents, lobbyists, and public officials
- material changes to project ownership or operation

Non-disclosure agreements should not prevent elected governments or residents from understanding the material financial, environmental, or infrastructural implications of a project.

The Province should also publish the results of its Strategic Priority Assessment and System Impact Assessment to the greatest extent reasonably possible.

Without public assessment records, communities will have difficulty determining why one project was approved over another, what commitments influenced the decision, and whether promised benefits

were ultimately delivered.

Recommendation: Create project-specific public records showing what was proposed, what was promised, what was approved and whether those commitments were met.

7. Ontario should define ‘digital sovereignty’

The draft Playbook appropriately identifies digital sovereignty as a strategic objective. It should plainly acknowledge that Canada cannot currently ensure sovereignty over its data in the cloud.

Data residency is only one component of digital sovereignty.

Assessment should also consider:

- ownership and control
- jurisdiction over operators and service providers
- access rights
- data exhaust
- reliance on foreign cloud platforms
- hardware and software supply chains
- cybersecurity
- contractual control
- the ultimate economic beneficiaries of public infrastructure decisions

A Canadian facility can still depend heavily on foreign technology and service providers, and fall under foreign legal jurisdictions. The Playbook must ensure effective control over legal access pathways, enforcement mechanisms, and decrypt-and-disclosure capabilities.

Ontario should therefore avoid treating "data stored in Ontario" as synonymous with "Ontario or Canadian control."

Recommendation: Develop a holistic digital-sovereignty assessment that considers ownership, control, legal jurisdiction, supply-chain dependence, and data governance.

8. Don’t approve first and regulate later

Ontario is still determining how these projects should be evaluated at the same time that municipalities are being approached with increasingly large proposals.

This creates pressure to make irreversible infrastructure and land-use decisions before the rules governing this new industrial category are complete.

The Province should therefore make sure the transition to the final Playbook does not create an incentive for proponents to accelerate projects through the regulatory gap.

Municipalities should retain the ability to adopt interim controls and establish local standards while provincial rules are finalized.

Recommendation: Preserve municipal authority to pause or condition development while local frameworks are being completed, and avoid creating grandfathering incentives for projects seeking approval before the final Playbook takes effect.

9. Clarify cooling-technology claims

Closed-loop cooling is a more water-efficient means of heat transfer.

While this reduces direct water consumption, it doesn't make a facility's consumption waterless or impact-neutral because the system still requires initial fluid, maintenance, and requires additional auxiliary electricity consumption.

In a regulatory environment where off-grid power generation is incentivized for data centres, closed-loop cooling, which itself requires more power, shifts water usage from the data centre to the power plant in an effective sleight-of-hand unless this is captured in the framework.

The US Department of Energy's Lawrence Berkeley National Laboratory found in 2023 that for every gallon of water used to cool American data centres, 12.4 gallons were used to cool their associated power stations.

Recommendation: Consider projects based on *total* measured water consumption, both within the data centre and any associated infrastructure, including power infrastructure.

10. Adopt leading-edge technology with forethought

We welcome the use of novel methods and technology to mitigate the ecological impacts of facilities. The Playbook suggests direct-to-chip liquid cooling where a cooling plate sits directly on high-heat components such as GPUs.

This is effective for modern AI chips, and allows more computing power in a smaller space. Higher GPU-density quickly cascades into more electricity consumption and more total heat generation.

Large datacentres operate mechanical equipment continuously. Quieter equipment, vibration isolation, waste-heat recovery, are all correct strategies. The Playbook must clarify how these technologies equate technical efficiency with environmental sustainability and community concerns.

Recommendation: Base approvals on absolute energy and resource consumption. Define actual noise levels residents will experience at their property boundaries with enforceable daytime and nighttime noise limits.

Conclusion

Ontario is right to develop a province-wide framework for data-centre development, but the emergence of this infrastructure is also exposing a larger governance problem.

Municipalities are closest to the physical consequences of these projects while often possessing the least information about their scale, infrastructural impacts and resource-needs, ownership, and long-term community impacts.

The Data Centre Playbook should close this regulatory gap.

Ontario can attract responsible digital infrastructure investment while requiring projects to demonstrate genuine public value, internalize their infrastructure costs, meet measurable environmental standards and obtain genuine support from the communities that will host them.